

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Contact cleaner, Part Number 3824510, or equivalent
- Lubriplate™ 105, Part Number 3163086, or equivalent

Additional Service Items

- Dial indicator
- Hydraulic press capable of 2224 N [5000 lbf]

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-018.html>)
- Remove the Low-Temperature Aftercooler (LTA) water pump support bracket. Refer to Procedure 008-195 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-195.html>)

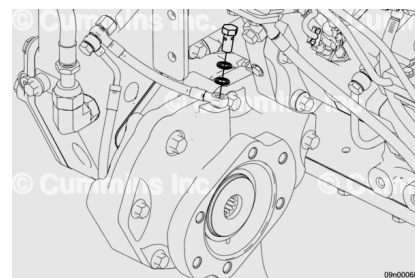
- Remove the LTA water pump. Refer to Procedure 008-109 in Section 8.
(/qs3/pubsys2/xml/en/procedures/57/57-008-109.html)

Remove

Accessory Drive Mounted

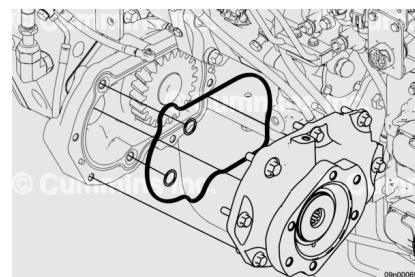
Disconnect lubricating oil supply line.

Remove and discard sealing washers.



Remove LTA water pump drive.

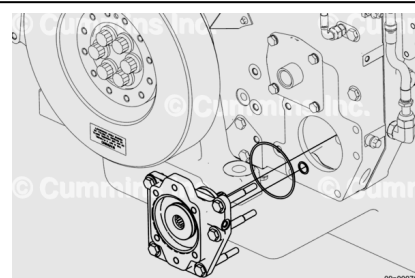
Remove and discard sealing o-rings and rectangular strip seal.



Front Gear Cover Mounted

Remove LTA water pump drive.

Remove and discard sealing o-ring and lubricating oil passage o-ring.



Clean and Inspect for Reuse

Accessory Drive Mounted

⚠ WARNING ⚠

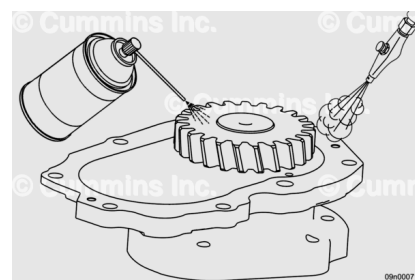
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

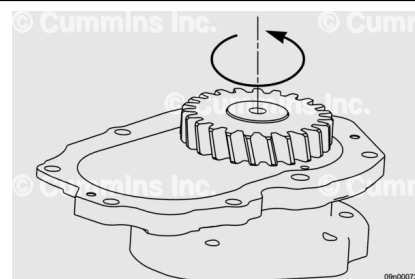
Clean the parts with contact cleaner, Part Number 3824510, or equivalent.

Dry the parts with compressed air.



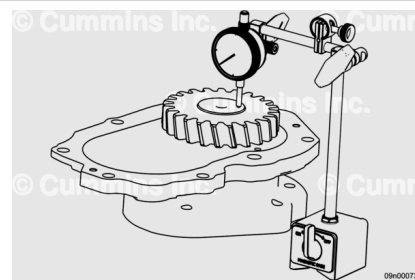
Turn gear of LTA water pump drive.

Rebuild LTA water pump drive if it does **not** turn smoothly.



Measure end clearance.

LTA Water Pump Drive End Clearance		
mm		in
0.19	MIN	0.0075
0.59	MAX	0.0232

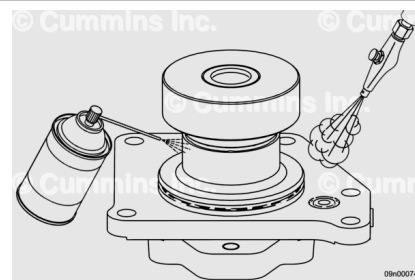


Replace thrust bearings if clearance is **not** within specifications.

Front Gear Cover Mounted

⚠ WARNING ⚠

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⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the parts with contact cleaner, Part Number 3824510, or equivalent.

Dry the parts with compressed air.

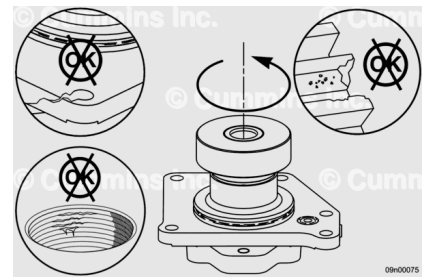
Inspect the LTA water pump drive for:

- Damaged or missing gear teeth
- Cracked housing
- Damaged capscrew threads
- Damaged sealing surfaces

Replace the LTA water pump drive if damage is found.

Check the rotation of the shaft. The shaft **must** turn smoothly and evenly.

Replace the LTA water pump drive if the shaft is difficult to turn or wobbles in the housing.

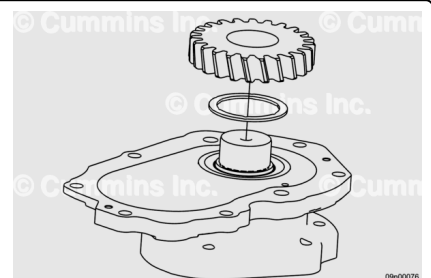


Disassemble

Accessory Drive Mounted

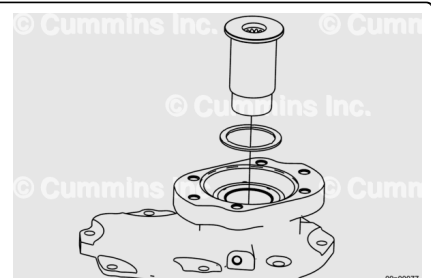
Use hydraulic press to remove gear from shaft assembly.

Remove thrust bearing.

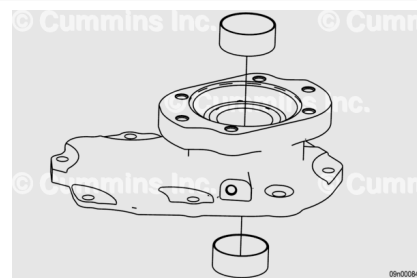


Remove shaft from LTA water pump drive.

Remove thrust bearing.



Use hydraulic press to remove bushings from LTA water pump drive.



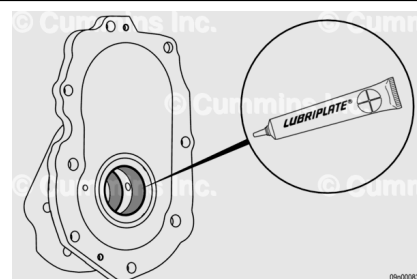
Assemble

Accessory Drive Mounted

Use hydraulic press to press bushings into LTA water pump drive.

Verify bushings are flush or no more than 0.2 mm [0.0079 in] below face.

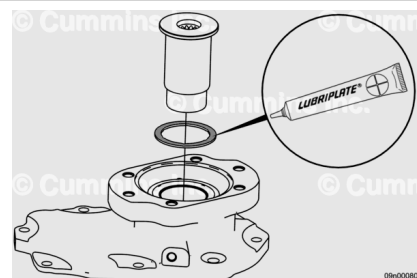
Lubricate bushings with Lubriplate™ 105, Part Number 3163086, or equivalent.



Lubricate bearing with Lubriplate™ 105, Part Number 3163086, or equivalent.

Place thrust bearing onto shaft.

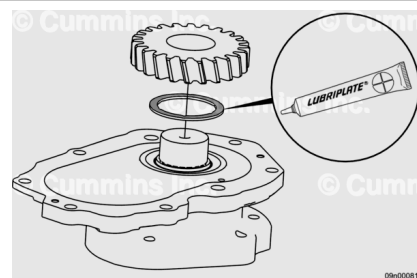
Install shaft into LTA water pump drive.



Lubricate bearing with Lubriplate™ 105, Part Number 3163086, or equivalent.

Place thrust bearing on end of shaft.

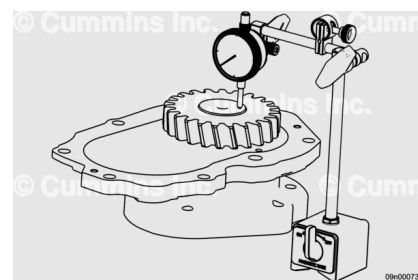
Use hydraulic press to press gear onto shaft until the gear contacts the shoulder of the shaft.



Measure end clearance.

LTA Water Pump Drive End Clearance

mm		in
0.19	MIN	0.0075
0.59	MAX	0.0232



Install

Accessory Drive Mounted

Install new sealing o-rings and rectangular strip seal.

Lubricate sealing o-rings with Lubriplate™ 105, Part Number 3163086, or equivalent.

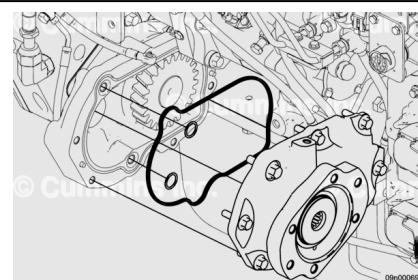
Install LTA water pump drive.

Torque Value:

Two M12 capscrews 80 n•m [60 ft-lb]

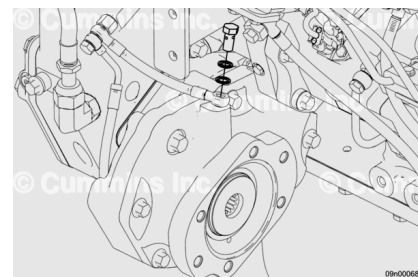
Torque Value:

Six M10 capscrews 46 n•m [34 ft-lb]



Connect lubricating oil supply line. Use new sealing washers.

Torque Value: 15 n•m [180 in-lb]



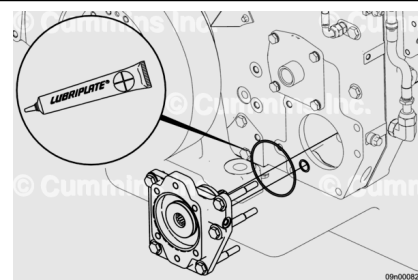
Front Gear Cover Mounted

Install new radial o-ring and lubricating oil passage o-ring.

Lubricate radial o-ring seal and lubricating oil passage o-ring with Lubriplate™ 105, Part Number 3163086, or equivalent.

Install LTA water pump drive.

Torque Value: 113 n•m [83 ft-lb]



Finishing Steps

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- Install the LTA water pump. Refer to Procedure 008-109 in Section 8.
(/qs3/pubsys2/xml/en/procedures/57/57-008-109.html)
- Install the LTA water pump support bracket. Refer to Procedure 008-195 in Section 8.
(/qs3/pubsys2/xml/en/procedures/57/57-008-195.html)
- Fill the LTA Cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/57/57-008-018.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate engine. Check for leaks.